GRAINGUARD · TRACEABILITY MATRIX

GrainGuard traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/grainguard		
Source	docs/03b-traceability.md @ 5c553ac		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	Generated by .kit/trace.py

GrainGuard traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 17 user requirements, 15 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	Grain temperature profile	GGD-CAL-001, GGD-BLD-001, GGD-STM-001, GGD-OPT-001	Analysis and inspection (GGD-CAL-001 C1, model), TRL 3	Met on paper
UR-002	PR-002	Grain moisture estimate	GGD-CAL-001, GGD-STM-001, GGD-OPT-001, BOM	Analysis (GGD-CAL-001 B5), TRL 3; comparison with oven or meter samples, TRL 4	At risk
UR-003	PR-003	Ambient air measurement	GGD-CAL-001, GGD-BLD-001, GGD-STM-001	Datasheet and shield error estimate; shield error test, TRL 4	To verify at TRL 4
UR-004	PR-004	Fan decision rule	GGD-CAL-001, GGD-BLD-001, BOM	Logic review (GGD-CAL-001 B10), TRL 3; season simulation, TRL 4	Met by design
UR-005	PR-005	Fail-safe control	GGD-CAL-001, GGD-BLD-001	Circuit review, TRL 3; bench test, TRL 4	Met by design
UR-006	PR-006	Heating alert	GGD-CAL-001, GGD-STM-001, GGD-OPT-001	Logic review (GGD-CAL-001 C4), TRL 3; heated-core test, TRL 4	At risk
UR-007, UR-017	PR-007	Radio link	GGD-CAL-001, GGD-BLD-001, GGD-STM-001	Link budget (GGD-CAL-001 E3 to E6), TRL 3; site survey, TRL 4	Met on paper
UR-008	PR-008	Power autonomy	GGD-CAL-001, BOM	Power budget (GGD-CAL-001 D5, D6, G5), TRL 3	Not met
UR-009	PR-009	Environment	GGD-CAL-001, GGD-BLD-001, GGD-STM-001	Datasheets and design review (GGD-CAL-001 G2 to G4), TRL 3; exposure test, TRL 4	Not met
UR-010	PR-010	Mechanical strength	GGD-CAL-001, GGD-BLD-001, GGD-STM-001, GGD-OPT-001	Force estimate (GGD-CAL-001 F2, F3), TRL 3; bin maker's data per bin	To verify at TRL 4
UR-011	PR-011	Install and service without grain entry	GGD-CAL-001, GGD-BLD-001	Installation sequence review (GGD-BLD-001), TRL 3	Met by design
UR-012	PR-012	Electrical isolation	GGD-CAL-001, GGD-BLD-001, GGD-STM-001, BOM	Circuit review and relay datasheet (GGD-CAL-001 D8), TRL 3	Met by design
UR-013	PR-013	Cost per bin	GGD-CAL-001	Priced BOM (`bom/bom.csv` ; GGD-CAL-001 H1), TRL 3	Over target
UR-014	PR-014	Local data	GGD-CAL-001, GGD-BLD-001	Architecture review, TRL 3	Met by design

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-015, UR-016	PR-015	Retrofit and workshop build	GGD-CAL-001, GGD-BLD-001	Inspection of the build plan (GGD-BLD-001) and the D3 model, TRL 3	Met by design

Gaps

None.